

SmartMOM Bot — API Reference & Integration Guide

A Complete Guide to Endpoint Lifecycles, Call Ordering, Specifications & Request/Response Contracts

Project: SmartMOM Bot (Meeting Intelligence Workspace)	Backend Stack: FastAPI, SQLAlchemy, SQLite/PostgreSQL
Auth Protocol: JWT Bearer Token (Authorization Header)	Core AI: Local Whisper CPP, Quantized Qwen2.5, DistilBERT
Base URL: /api (Proxied via Nginx / Vite)	Document Version: 1.0.0 (Production Verified)

1. Recommended Endpoint Execution Sequence

To achieve a complete automated meeting lifecycle—from raw spoken audio to verified, downloadable minutes and analytics—endpoints should be executed in the exact ordered sequence detailed below:

Step	Endpoint & Method	Lifecycle Phase	Meeting Status	Role Required
0	GET /health	System Pre-flight check	N/A	Public
1	POST /api/auth/register or /api/auth/login	Authentication & JWT retrieval	N/A	Public
2	GET /api/auth/me	Session validation & profile load	N/A	Authenticated
3	POST /api/meetings	Upload audio recording	uploaded	Organizer
4	POST /api/meetings/{id}/participants	Invite registered participants	uploaded	Organizer
5	POST /api/meetings/{id}/transcribe	Whisper Speech-to-Text extraction	transcribing → transcribed	Organizer
5a	PUT /api/meetings/{id}/transcript	Optional manual transcript edits	transcribed	Organizer
6	POST /api/meetings/{id}/analyze	AI Minutes & Sentiment Analysis	ready	Organizer
6a	PUT /api/meetings/{id}/summary	Human review & edits (versioned)	saved	Organizer
7	GET /api/meetings/{id}/export.pdf	Download formal PDF minutes	ready / saved	All Members
8	POST /api/meetings/{id}/feedback	Submit 1–5 star rating & feedback	ready / saved	All Members
9	DELETE /api/meetings/{id}	Permanent cleanup & file deletion	Deleted	Organizer

Why this order matters:

- **Audio Upload before Transcription:** Transcribing requires a valid physical audio file on disk linked to the database record. Calling /transcribe without an audio payload triggers HTTP 400.
- **Transcription before Analysis:** The AI minutes generator (Qwen2.5) and sentiment analyzer (DistilBERT) require non-empty text input. Calling /analyze before /transcribe returns HTTP 400.
- **Analysis before Summary Editing:** The summary editor allows organizers to refine AI-extracted points. Saving a summary (PUT /summary) creates a snapshot in meeting_versions.
- **Finalization before PDF Export:** Exporting to PDF pulls directly from meeting.summary; generating the summary first ensures a rich, multi-section PDF document.

2. Authentication & User Profile Endpoints

Authentication uses JSON Web Tokens (JWT) signed with HMAC-SHA256. Tokens are transmitted in the HTTP Authorization: Bearer <token> header.

POST	User Registration /api/auth/register	Role: Public
Purpose	Registers a new Organizer or Participant user with hashed password and returns a JWT session token.	
Request	JSON: {"name": "Kalsoom Khan", "email": "kalsoom@fyp.test", "password": "Password@123", "role": "organizer"}	
Response	201 Created: {"token": "eyJhbGciOiJIUzI1Ni...", "user": {"id": "usr_01", "name": "Kalsoom Khan", "email": "kalsoom@fyp.test", "role": "organizer"}}	

POST	User Authentication /api/auth/login	Role: Public
Purpose	Verifies credentials against stored password hash (Argon2/bcrypt) and returns a valid JWT session token.	
Request	JSON: {"email": "admin@smartmom.test", "password": "Admin@12345"}	
Response	200 OK: {"token": "eyJhbGciOiJIUzI1Ni...", "user": {"id": "usr_admin", "name": "Admin", "email": "admin@smartmom.test", "role": "organizer"}}	

GET	Current User Profile /api/auth/me	Role: Authenticated User
Purpose	Validates the incoming Bearer token and returns identity and role metadata for the current session.	
Request	Headers: Authorization: Bearer <token>	
Response	200 OK: {"id": "usr_admin", "name": "Admin", "email": "admin@smartmom.test", "role": "organizer"}	

PUT	Update Account Role /api/auth/me/role	Role: Authenticated User
Purpose	Updates the user's default role between 'organizer' and 'participant'.	
Request	JSON: {"role": "participant"}	
Response	200 OK: {"id": "usr_admin", "name": "Admin", "email": "admin@smartmom.test", "role": "participant"}	

3. Meeting Creation & Collaboration Endpoints

GET	List Accessible Meetings /api/meetings	Role: Authenticated User
Purpose	Returns all meetings owned by or shared with the authenticated user, ordered from newest to oldest.	
Request	Headers: Authorization: Bearer <token>	
Response	200 OK: [{"id": "mtg_123", "title": "FYP Review", "status": "ready", "access_role": "organizer", "created_at": "2026-09-04T10:00:00Z", ...}]	

POST	Create Meeting & Upload Audio /api/meetings	Role: Organizer / User
Purpose	Accepts multi-part audio file (WAV, MP3, M4A, OGG, WebM up to 200MB) and stores it safely under uploads/.	
Request	multipart/form-data: audio=<binary_file> (required), title="Client Kickoff" (optional)	
Response	201 Created: {"id": "mtg_abc123", "title": "Client Kickoff", "status": "uploaded", "access_role": "organizer"}	

GET	List Meeting Members /api/meetings/{id}/participants	Role: Organizer or Participant
Purpose	Retrieves the list of all users who have access to this meeting and their granted role.	
Request	URL Param: id (string) Headers: Authorization: Bearer <token>	
Response	200 OK: [{"id": "u1", "name": "Organizer", "email": "admin@test.com", "access_role": "organizer"}, {"id": "u2", "name": "Participant", "email": "part@test.com", "access_role": "participant"}]	

POST	Invite Participants <code>/api/meetings/{id}/participants</code>	Role: Organizer Only
Purpose	Adds registered users to the meeting by email address so they can view summaries and submit feedback.	
Request	JSON: <code>{"emails": ["part1@test.com", "part2@test.com"]}</code> (or comma-separated in "email")	
Response	201 Created: <code>{"invited": [{"id": "u2", "email": "part1@test.com", ...}], "missing": ["part2@test.com"]}</code>	

4. AI Transcription, Analysis & Versioning Endpoints

POST	Run Speech-to-Text Transcription <code>/api/meetings/{id}/transcribe</code>	Role: Organizer Only
Purpose	Transcribes audio using local Whisper CPP (small.en-tdrz) with speaker-turn diarization into Speaker 1/2 segments.	
Request	URL Param: id (string) Headers: Authorization: Bearer <token>	
Response	200 OK: <code>{"text": "Speaker 1: Welcome...\nSpeaker 2: I reviewed the tasks.", "segments": [{"speaker": "Speaker 1", "text": "Welcome..."}], "mode": "local-whisper"}</code>	

PUT	Manual Transcript Correction <code>/api/meetings/{id}/transcript</code>	Role: Organizer Only
Purpose	Allows organizer to edit or replace transcript text before running downstream AI intelligence pipelines.	
Request	JSON: <code>{"transcript": "Speaker 1: Let's start the project demo.\nSpeaker 2: Documentation is completed."}</code>	
Response	204 No Content	

POST	Extract Minutes & Analyze Sentiment <code>/api/meetings/{id}/analyze</code>	Role: Organizer Only
Purpose	Executes dual AI pipelines: Quantized Qwen2.5 for structured minutes (agenda, decisions, actions) and DistilBERT for sentiment & engagement share.	
Request	URL Param: id (string) Headers: Authorization: Bearer <token>	
Response	200 OK: <code>{"summary": {"agenda": ["Review code"], "decisions": ["Merge branch"], "discussion": ["All tests passing"], "actions": [{"owner": "Kalsoom", "task": "Release candidate", "due": "Friday"}]}, "analysis": {"overall": "positive", "participants": [{"name": "Speaker 1", "sentiment": "Positive", "engagement": 54}]}, "mode": "local-qwen"}</code>	

PUT	Save Refined Minutes & Version <code>/api/meetings/{id}/summary</code>	Role: Organizer Only
Purpose	Saves human edits to agenda, decisions, discussion notes, and action items. Automatically appends a snapshot in meeting_versions for audit compliance.	
Request	JSON: <code>{"summary": {"agenda": ["Project final signoff"], "decisions": ["Approved"], "discussion": [...], "actions": [{"owner": "Lead", "task": "Deploy to Docker", "due": "Monday"]}}</code>	
Response	200 OK: <code>{"summary": {"agenda": ["Project final signoff"], ...}}</code>	

5. Export, Feedback & Cleanup Endpoints

GET	Export Formal PDF Minutes <code>/api/meetings/{id}/export.pdf</code>	Role: Organizer or Participant
Purpose	Dynamically generates and streams an A4 PDF document with clean headers, agenda, decisions, discussion notes, and assigned action item table.	
Request	URL Param: id (string) Headers: Authorization: Bearer <token>	
Response	200 OK (Binary Stream: application/pdf) Content-Disposition: attachment; filename="FYP-Review-minutes.pdf"	

POST	Submit Meeting Feedback <code>/api/meetings/{id}/feedback</code>	Role: Organizer or Participant
Purpose	Stores quality ratings (1 to 5 stars) and qualitative feedback comments from any authorized attendee.	
Request	JSON: <code>{"rating": 5, "comment": "Accurate action items, loved the sentiment analysis breakdown!"}</code>	
Response	201 Created: <code>{"ok": true}</code>	

DELETE	Delete Meeting & Audio File /api/meetings/{id}	Role: Organizer Only
Purpose	Cascades deletion through database (members, versions, feedback) and permanently removes audio file from disk.	
Request	URL Param: id (string) Headers: Authorization: Bearer <token>	
Response	204 No Content	

6. Role-Based Access Control (RBAC) Matrix

The table below clarifies endpoint permissions based on user status and meeting role.

Endpoint	Method	Anonymous / Public	Participant	Meeting Organizer
/health	GET	Allowed	Allowed	Allowed
/api/auth/register	POST	Allowed	N/A	N/A
/api/auth/login	POST	Allowed	N/A	N/A
/api/auth/me	GET	Denied (401)	Allowed	Allowed
/api/meetings	GET	Denied (401)	Allowed	Allowed
/api/meetings	POST	Denied (401)	Allowed (Becomes Owner)	Allowed
/api/meetings/{id}/participants	GET	Denied (401)	Allowed (If Member)	Allowed
/api/meetings/{id}/participants	POST	Denied (401)	Denied (404/403)	Allowed
/api/meetings/{id}/transcribe	POST	Denied (401)	Denied (404)	Allowed
/api/meetings/{id}/transcript	PUT	Denied (401)	Denied (404)	Allowed
/api/meetings/{id}/analyze	POST	Denied (401)	Denied (404)	Allowed
/api/meetings/{id}/summary	PUT	Denied (401)	Denied (404)	Allowed
/api/meetings/{id}/export.pdf	GET	Denied (401)	Allowed (If Member)	Allowed
/api/meetings/{id}/feedback	POST	Denied (401)	Allowed (If Member)	Allowed
/api/meetings/{id}	DELETE	Denied (401)	Denied (404)	Allowed

7. Standard Error Codes & Troubleshooting

All error responses return a standardized JSON structure: {"error": "Descriptive message"}.

Status Code	Condition / Cause	Remediation / Resolution
400 Bad Request	Unsupported audio MIME, missing email for invitation, or empty transcript submitted for analysis.	Ensure audio is WAV/MP3/M4A/OGG/WebM; verify transcript is populated prior to analyze.
401 Unauthorized	Missing Authorization header, invalid token, or expired session.	Call /api/auth/login to get a fresh token and pass as 'Bearer <token>'.
404 Not Found	Meeting does not exist, or current user lacks required access role.	Verify meeting ID and ensure user has organizer role for modification requests.
409 Conflict	Attempted registration with an email that is already registered.	Use /api/auth/login or sign up with an alternative email address.
413 Payload Too Large	Audio file exceeds MAX_UPLOAD_MB threshold (default 200 MB).	Compress audio to mono 16kHz WAV or 64kbps MP3 prior to uploading.
503 Unavailable	Local Whisper CLI / model files missing, or AI provider unconfigured.	Run scripts/setup-local-whisper.ps1 or verify OPENAI_API_KEY if in cloud mode.